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Learning to be rational or irrational?Explore the decision consistency between human beings and AI Agent in choice behaviour †

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Abstract: Since the advent of Large Language Models (LLMs), Generative AI has been extensively applied in the marketing field to improve decision-making efficiency and automate content generation. However, the effectiveness of AI Agent in assisting or substituting for human decisions remains controversial. This research conducted two experiments and trained Generative AI to perform decision-making tasks with different option-framing, aiming to evaluate the decision consistency between LLMs and humans. Results reveal that while LLMs can align with human decision patterns under certain conditions, they also exhibit irrational decision biases.

Keywords: *Generative AI, AI Agent, decision-making system, Option framing, decision bias*

1. Introduction

With the widespread application of Generative Artificial Intelligence and Large Language Models (LLMs) such as GPT-4, AI has gradually permeated areas such as everyday decision-making support, recommendation systems, and marketing strategy design. AI is increasingly expected to simulate, or even replace, human judgment when facing complex choices. However, while these technologies can enhance efficiency, they also raise significant concerns regarding bias and fairness.

According to a 2024 BBC report, AI tools used for resume screening and interview prediction may unintentionally filter out qualified candidates, resulting in the loss of top talent [1]. These cases show that when AI mimics human decision-making, it can reproduce and even amplify social biases. Although such tools offer efficiency, their increasing role in decision-making raises serious concerns. Public awareness of AI bias is growing, but most research still focuses on algorithmic detection or case studies, leaving a gap in understanding whether AI shares human-like cognitive biases in real-world contexts.

Human decisions are shaped by subtle cues such as framing, default options, and attribute order—often interacting with bounded rationality [2]. For instance, in aviation, how ticket features like baggage are framed—whether included or as add-ons can sway customer choices [3]. These effects highlight how presentation influences judgment, making them useful for exploring bias in AI. Another BBC report notes that 1 in 10 Britons

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have used AI to plan vacations, providing a real-life setting to compare human and AI decision behavior [4].

To address this gap, the overarching aim of this study is to explore whether AI agents such as GPT-3.5 and GPT-4 exhibit human-like decision biases when placed in marketing-inspired scenarios involving option framing and product attribute variations. By designing a task that mirrors real-world decision-making contexts, the study examines whether these language models respond similarly to humans under manipulations such as additive versus subtractive framing and default option effects. In doing so, the research bridges theories from marketing psychology and AI bias, offering empirical insights into whether GPT's outputs reflect genuine cognitive distortions or merely statistical patterns derived from training data. This contributes to a deeper understanding of AI explainability, reliability, and its alignment with human judgment in complex decision environments.

This study examines how additive and subtractive framing in a flight service selection task affects AI decision-making using GPT-3.5 and GPT-4. The Results section reports significant differences in decision time, number of services selected, and preference patterns. The Discussion highlights framing-induced bias, while the Conclusion underscores the need for transparent and context-aware AI design.

2. Materials and Methods

This research aims to investigate whether Large Language Model (LLM)-based AI Agents exhibit irrational decision-making behaviors akin to human when selecting in-flight services, alongside an examination of the consistency between its decision profiles and human ones. A behavioral experiment was conceived for this purpose, in this experiment, participants and AI Agents perform in-flight service selection tasks under two distinct option framing: Additive and Subtractive.

In order to comprehensively assess the convergences and divergences in decision-making between AI Agents and humans, the study focuses on three indicators: (1) decision time required to complete the task (2) the final quantity of services retained and (3) the propensity towards either hedonic or utilitarian service types. In this chapter, we will explain the experimental design, how data was collected and preprocessed, and the statistical methods used.

2.1. Experimental Design

This study employed an in-flight service selection task, simulating long-haul flight purchases, to evaluate AI-Agent decision-making. The experiment featured 12 modifiable ancillary services, categorized as either hedonic or utilitarian. To explore framing effects on decision processes, two conditions were implemented: Additive Framing (+OF), where models incrementally selected services from a null set; and Subtractive Framing (–OF), where models deselected from a pre-selected full set. These frameworks maintained identical service and pricing information, ensuring equivalence. 2×2 factorial design, combining Model Type (GPT-3.5, GPT-4) and Option Framework (+OF, –OF), generated four experimental conditions. Under each condition, language models performed selections/deselections on the same 12 services, enabling a comparative analysis of model and framework impacts on decision outcomes against human behavioral patterns.

2.2 AI-Agent

This study simulated AI-Agent decision-making behavior in actual consumer scenarios by designing systematic prompts that provided the AI-Agents with the option framing context (additive or subtractive), in-flight service items, and simulated human characteristics (gender, age, flight frequency). For each simulation, the AI-Agent's response was

recorded and converted into a structured data table, simultaneously calculating the following metrics: decision time, quantity of services retained, and service attribute distribution. Finally, the AI data were integrated with real human questionnaire data for analysis and comparison, with the specific process illustrated in Figure 2.2.1.

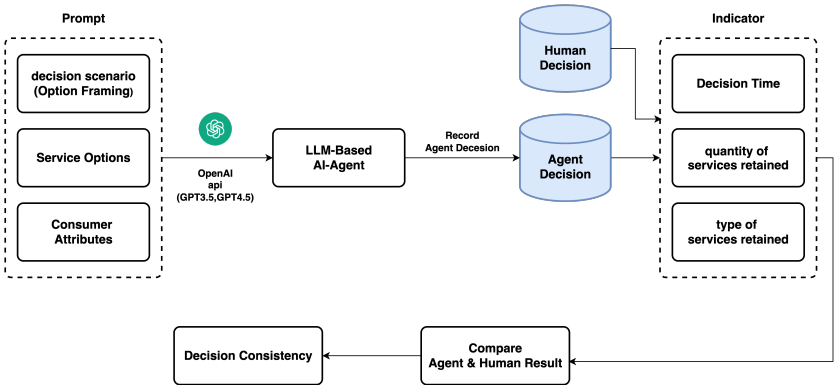


Figure2.2.1. AI-Agent Design

2.3 Participants / Data Sources

This study collected data from two distinct sources: survey responses from real human participants and simulated data generated by LLM-Based AI-Agents Human data was gathered from 140 valid questionnaires administered at Taiwan Taoyuan International Airport over two weeks, covering diverse demographics (52% female, 48% male; 60% aged 21-40; varied flight frequency and purpose).

AI Agent data was generated through a prompt-based methodology, where LLMs simulated decision processes under experimental conditions and personal characteristics identical to those for humans. The consistency of prompts across both data collection methods ensured a robust comparative foundation for the study.

2.4 Data Preprocess

To ensure the quality and consistency of AI-Agent simulated data, GPT model outputs underwent pre-processing. This involved data cleaning to exclude samples with formatting errors, unreasonable decisions (e.g., all 0 or 12 items retained), and outlier decision times (based on Z-scores). Valid outputs were then normalized into three metrics: Decision Time (seconds), Total Services Retained (0-12 items), and Proportion of Hedonic Services (%).

2.5 Statistical/AI-Agent Methods

To examine the differences in AI-Agent decision-making behavior under varying option frameworks (additive vs. subtractive), this study conducted statistical analyses primarily from three dimensions. Appropriate analytical methods were employed based on the characteristics of each variable, as detailed below:

For Decision Time and Service Attribute Preference, the analysis method consistently involved both independent samples t-tests and Mann–Whitney U tests. For Fairness of Decision Distribution, assessed via Shannon Entropy, descriptive statistics and intergroup comparisons were employed to ascertain the concentration or dispersion of preferences.The significance level was set at α=0.05; if the p-value was less than 0.01, it was considered highly significant.

All statistical analyses were conducted in a Python 3.10 environment. AI-Agent data were simulated using a prompt-based approach via the OpenAI API, leveraging GPT-3.5 and GPT-4 models to mimic decision-making under varying option framing conditions.

3. Results

To explore whether AI exhibits human-like rational and irrational behavior in decision-making, this chapter presents results from three levels of analysis. First, we examine whether AI is sensitive to framing effects, as humans are. Second, we assess bias and fairness across different AI models by evaluating their tendency to show systematic preferences. Finally, we compare AI choices with those of human participants under the same conditions to evaluate the consistency of AI in simulating human decision-making.

3.1. Impact of Framework Types on AI Decision Making Behavior

Human decision-making is often influenced by contextual factors and problem framing, resulting in shifts in choice tendencies—commonly known as the framing effect. In this section, we manipulate additive and subtractive frames to examine whether AI systems exhibit similar context-dependent, potentially irrational response patterns. By analyzing both choice outcomes and reaction times, we assess the extent to which AI is sensitive to contextual changes and the consistency of its decision-making behavior.

3.1.1. Decision Time Comparison

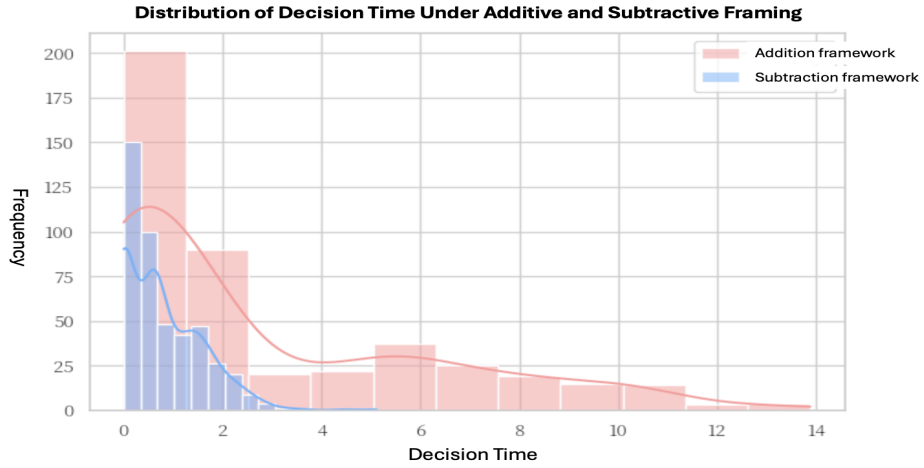
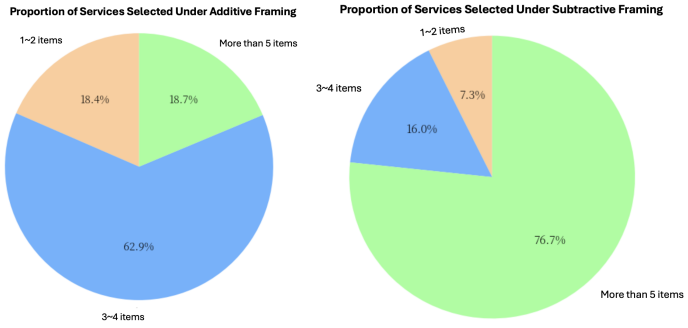


Figure3.1.1. Distribution of Decision Time Under Additive and Subtractive Framing

As illustrated in Fig. 3.1.1.1., the average decision time of the AI in the additive frame is significantly longer than in the subtractive frame, with some responses exceeding 10 seconds. In contrast, decision times under the subtractive frame are predominantly concentrated within 1 second, suggesting that the model responds more rapidly when processing tasks in this context.

Statistical results show a significant effect of the starting point (full-empty vs. full-full) on AI response speed ($t = 12.556$, $p < 0.001$; $U = 131180.5$, $p < 0.001$).

3.1.2. Comparison of the Number of Services Retained



(a) (b)

Figure 2. (a) Proportion of services selected under additive framing. Most participants selected 3–4 items (62.9%), while fewer chose 1–2 items (18.4%) or more than 5 items (18.7%);
(b) Proportion of services selected under subtractive framing. A higher proportion of participants chose more than 5 items (76.7%) compared to 3–4 items (16.0%) and 1–2 items (7.3%)

The graph illustrates the distribution of the number of services selected by the AI under the two framing conditions. In the additive frame, the AI predominantly chooses 3 to 4 services (62.9%), with the distribution exhibiting a clear central tendency, indicating a moderate preference strategy. In contrast, under the subtractive frame, the AI shows a strong inclination to retain most services, with 76.7% of instances involving the selection of more than five items. Th**is sugg**ests that the model is less likely to remove a large number of options when the initial state is "select all".

A significant difference in the number of selected services was found between conditions ($t = -21.863$, $p < 0.001$; $U = 32{,}369.5$, $p < 0.001$).

3.1.3 Service attribute preference analysis

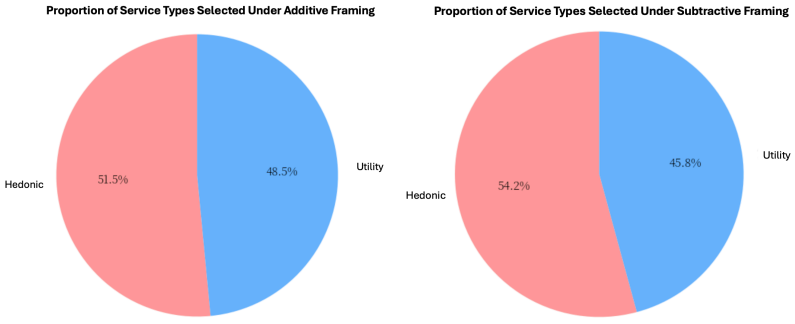


Figure 3.1.3.1. Service attribute preference under different option framings.
(a) Under additive framing, the proportion of selected hedonic services (51.5%) was slightly higher than utility services (48.5%), suggesting a near-balanced preference.
(b) Under subtractive framing, hedonic services were more frequently retained (54.2%) than utility services (45.8%), indicating a shift toward hedonic preference when starting from a full selection set.

As shown in the graph, the proportion of Hedonic and Utility services selected by the AI remains relatively stable across both framing conditions, with a slight overall preference for Hedonic options. In the additive frame, 51.5% of the selected services are Hedonic and 48.5% are Utility; in the subtractive frame, the proportions are 54.2% and 45.8%, respectively. This indicates that the framing effect has minimal impact on the AI's tendency to favor one type of service over the other.

3.2. Decision Bias and Fairness Analysis of AI Models

3.2.1. Bias analysis in an additive framework (+OF)

Table3.2.1.1. Entropy Values for Fairness Under Additive Framing by Model

GPT-4	GPT-3.5	Random
3.3395	3.2962	3.5749

The stochastic model exhibits the highest entropy value, reflecting a more evenly distributed pattern of choices, which aligns with theoretical expectations. In contrast, the GPT models display

GPT-4 and GPT-3.5 exhibit significantly higher selection proportions for certain service items, reflecting a more uneven distribution of choices. Specifically, GPT-4 demonstrates a strong preference for Additional Baggage and In-flight Internet, whereas GPT-

3.5 shows a marked inclination toward Priority Check-in and Lounge. In contrast, the stochastic model displays relatively uniform selection proportions across all service items, indicating no distinct preference patterns.

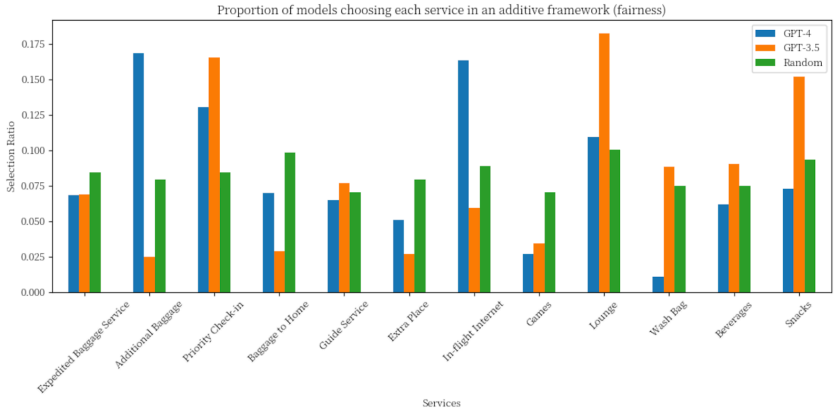


Figure 3.2.1.1. Proportion of models choosing each service in an additive framework(fairness)

GPT-4 and GPT-3.5 exhibit significantly higher selection proportions for certain service items, reflecting a more uneven distribution of choices. Specifically, GPT-4 demonstrates a strong preference for Additional Baggage and In-flight Internet, whereas GPT-3.5 shows a marked inclination toward Priority Check-in and Lounge. In contrast, the stochastic model displays relatively uniform selection proportions across all service items, indicating no distinct preference patterns.

3.2.2. Bias Analysis in a Subtractive Framework (−OF)

Table3.2.2.1. Entropy Values for Fairness Under Subtraction Framing by Model l

GPT-4	GPT-3.5	Random
3.3261	3.474	3.584

GPT-4 exhibits the lowest entropy among the three models, indicating the highest degree of preference concentration under the subtractive frame. GPT-3.5 shows a comparatively higher entropy, reflecting a more dispersed choice pattern, yet still lower than that of the stochastic model. This suggests that GPT-3.5, while less biased than GPT-4, still demonstrates a certain degree of preference tendency.

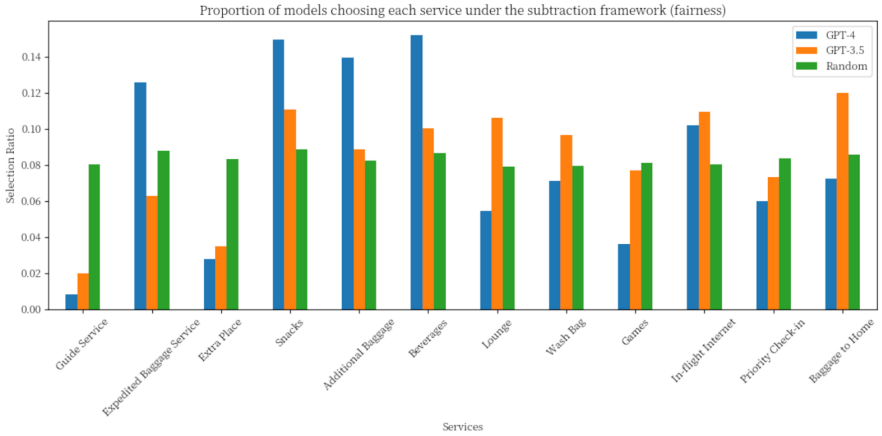


Figure 3.2.2.1. Proportion of models choosing each service in an additive framework(fairness)

GPT-4 exhibits a significantly higher selection proportion for multiple service items, such as Snacks, Beverages, and Additional Baggage, indicating a clear preference pattern. In contrast, GPT-3.5 demonstrates stronger preferences for Baggage to Home, Lounge, and Wash Bag. Meanwhile, the stochastic model (green) displays a relatively uniform distribution of selections across all options, reflecting the absence of pronounced preference tendencies.

3.3. Decision Bias and Fairness Analysis of AI Models

Furthermore, we compare the AI's choices with those of human participants under the same framing conditions to evaluate the consistency of AI behavior in simulating human decision-making. By examining changes in the number of items selected in the additive and subtractive scenarios, we assess whether the AI exhibits response patterns similar to those of human subjects. This comparison enables a more precise evaluation of the behavioral alignment between AI and humans, particularly in terms of framing effects and decision-making tendencies.

Table3.2.1.1. Entropy Values for Fairness Under Additive Framing by Model

	GPT-4	**GPT-3.5**
Mean	-0.58	-2.25
Standard Deviation	0.37	0.43
Sample Size (n)	3	3
t-value	11.22	3.01
p-value (two-tailed)	**0.007*** **	0.09

[1] p<0.05*, p<0.01***

We conducted a statistical analysis to compare AI and human choice behavior. Using the mean difference, standard deviation (SD), and a sample size of 3, we performed two-tailed t-tests to assess significance. GPT-4 showed a mean difference of –0.58 (SD = 0.37, t = 11.22, p = 0.007), indicating a strong similarity to human choices. In contrast, GPT-3.5 had a larger mean difference of –2.25 (SD = 0.43, t = 3.01, p = 0.09), suggesting it overestimated the number of choices and aligned less closely with human behavior.

4. Discussion

4.1 Framing Effects and Bias Generation Mechanisms in Language Models

This study reveals that additive (+OF) and subtractive (−OF) framing significantly affect language model decision-making, mirroring human cognitive biases. In subtractive conditions, models retain more options and favor hedonic services, reflecting human status quo bias and loss aversion (Tversky & Kahneman, 1981).

Despite lacking emotional reasoning, models exhibit contextually-driven behavior, particularly retaining semantically salient options from training data in subtractive tasks. This manifests as disproportionate retention of hedonic services (comfort, entertainment) even when options are functionally equivalent, indicating corpus-driven affective salience as a latent selection bias driver. Under subtractive framing, emotionally charged options become "sticky" due to frequent positive associations in training data, creating asymmetric selection patterns that mirror human affective biases.

The findings validate that semantically equivalent operations with different initial states (blank vs. fully selected) systematically alter model behavior, demonstrating susceptibility to framing manipulation and highlighting how corpus structure shapes framing-induced preferences.

4.2 Model Version and Contextual Sensitivity Interactions

Results reveal highly context-sensitive behavioral differences across model versions, displaying "model version × framing type" interactions rather than linear progression. GPT-3.5 shows higher bias in additive tasks (lower entropy, concentrated selections) but more dispersed, fairer behavior in subtractive tasks. Conversely, GPT-4 demonstrates uniform selections in additive tasks (higher entropy, lower bias) but pronounced preference concentration in subtractive tasks, amplifying bias.

This pattern indicates that bias generation does not improve unidirectionally with model capability but may intensify due to enhanced contextual sensitivity, leading to "contextual overfitting." These findings warn that improved linguistic accuracy does not automatically reduce bias—stronger models' heightened "contextual response intensity" must be incorporated into bias risk assessment to prevent high-performance models from amplifying contextual bias.

4.3 Human-Model Choice Behavior Comparison

Direct comparison reveals GPT-4's service selection patterns in additive conditions closely match human behavior ($p > 0.05$), while GPT-3.5 shows directionally consistent but less stable performance with larger human differences. Results indicate models can approximate human decision patterns under specific contextual guidance despite lacking genuine reasoning structures. However, this consistency remains context-dependent and unstable, particularly with amplified bias in complex or subtractive tasks, suggesting controlled-condition alignment without human-like reasoning mechanisms.

4.4 Limitations and Future Directions

This study validates language model bias formation under contextual manipulation and empirically compares human-model choice consistency. Findings reveal that model bias stems not solely from models themselves but from task context and prompt design interactions.

Limitations include: (1) single-category context (flight service selection) with structured attributes, excluding ambiguous, multi-turn, or moral judgment tasks; (2) limited human sample size using only choice outcomes without subjective variables like decision motivation or risk perception.

Future research should examine context designers' ethical responsibilities, establish prompt design transparency standards, and employ methods like eye-tracking or decision confidence assessment to explore model-human cognitive similarities. Cross-architectural comparisons (Claude, Gemini, etc.) could establish decision behavior bias maps, providing deeper insights into generative model controllability and bias distribution in complex decision tasks.

5. Conclusion

This study demonstrates that large language models inherit both rational and irrational decision patterns from human training data, producing context-dependent choice behaviors. GPT-4 exhibits human-like distributions in additive tasks but shows corpus-driven retention bias in subtractive contexts, despite semantically equivalent options. This bias systematically favors hedonic over utilitarian services, revealing affective skew in model outputs aligned with emotionally salient representations in training corpora.

Enhanced model capabilities do not eliminate biases but amplify contextual sensitivity and semantic overfitting. GPT-4's retention of salient options in subtractive tasks reflects generation mechanisms guided by corpus distributions, compromising choice fairness through interactions between context design and training data.

For sensitive applications, contextual prompt design constitutes a critical bias control point. We recommend prioritizing additive framing structures and establishing auditable mechanisms for prompts and generation processes. Developing cross-model benchmarks and response curves is essential for ensuring AI decision-making reliability and ethical accountability.

Supplementary Materials: Not applicable.

Author Contributions:
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Abbreviations

Abbreviation	Full Term
AI	Artificial Intelligence
LLMs	Large Language Models
GPT	Generative Pre-trained Transformer
AI Agent	Artificial Intelligence Agent
OF	Option Framing
+OF	Additive Option Framing
−OF	Subtractive Option Framing

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